LAB: 3

Task 1:

Assume that the following varriables are defined

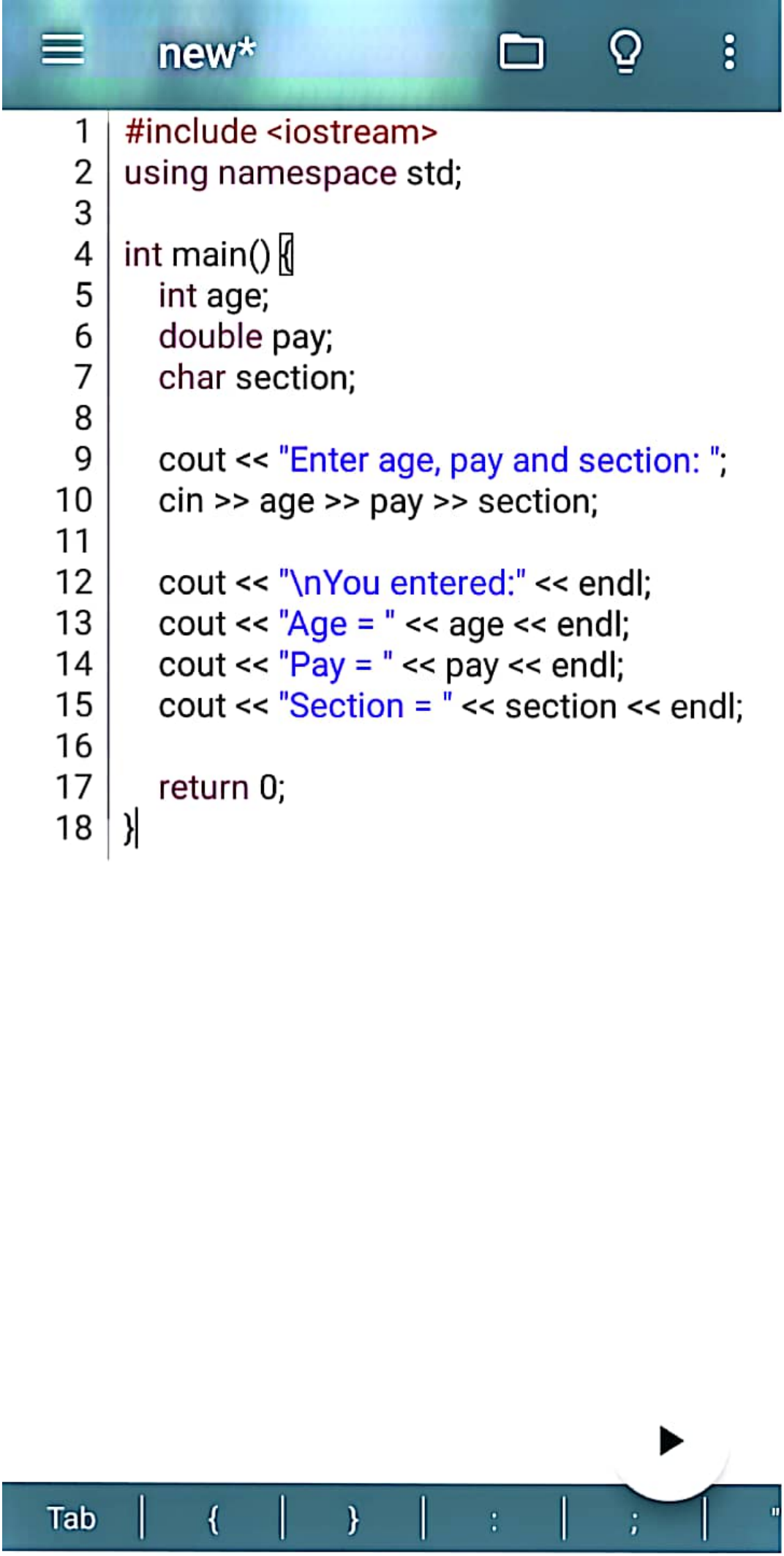
```
int age;
double pay;
char section;
```

```
#include <iostream>
using namespace std;
int main () {
    int age;
    double pay;
    char section;
        cout<< "Enterage, Pay and section:";
        cin>> age >> pay >>section;
    cout<<"\nyou entered:"<< endl;
    cout << "Age =" << age << endl;
    cout <<"pay =" << pay <<endl;
    cout << "section = "<< section <<endl;
    return 0;
}
```

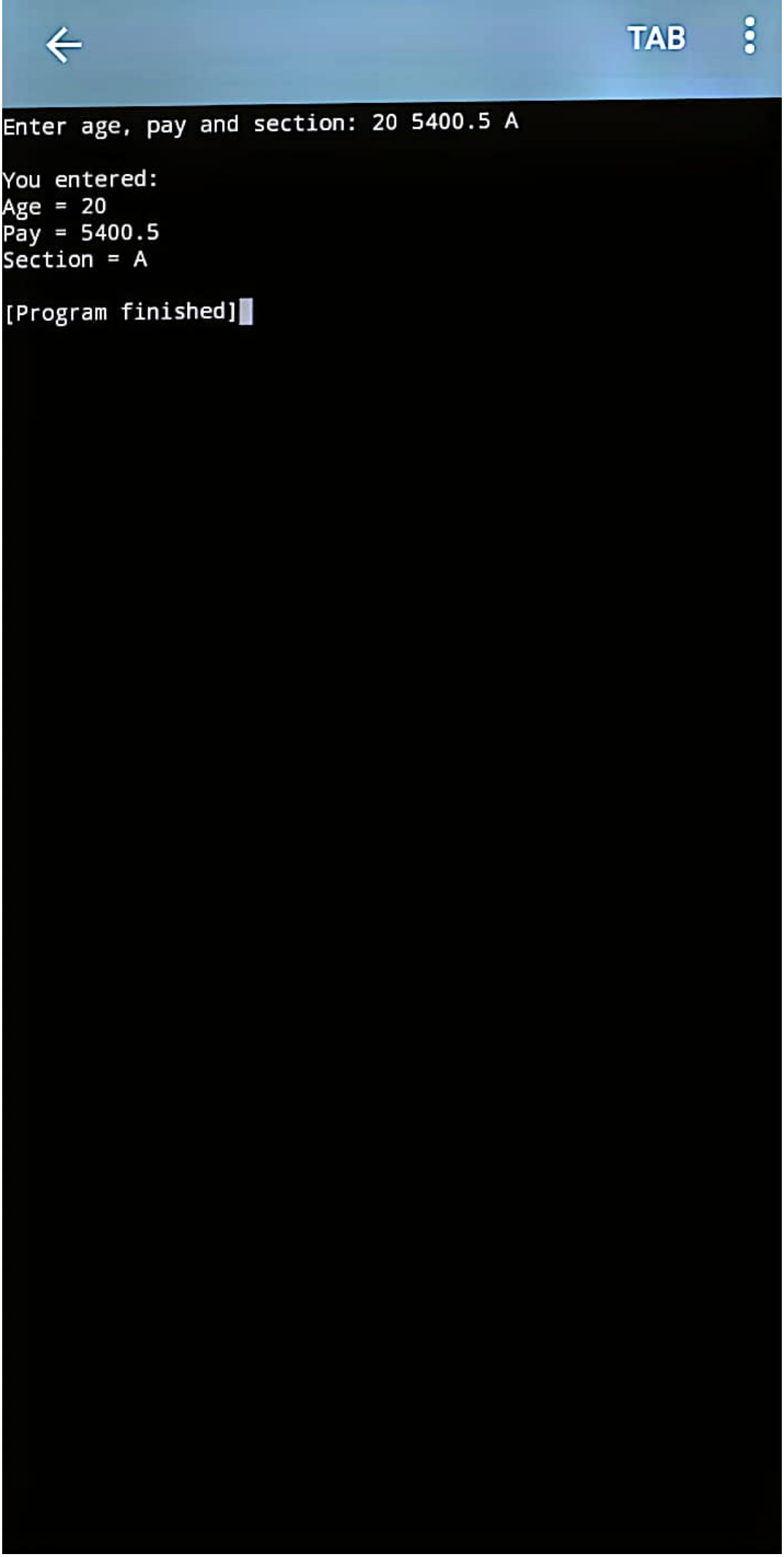
outPut: Enter age, Pay and section:

```
Age = 20
Pay = 54000.5
Section = A
```

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Task: 2

```
#include <iostream>
using namespace std;
int main() {
    cout << 28/4-2<<endl; //5
    cout << 6+12*2-8<<endl;      //22
    cout << 4+8*2 <<endl;        //20
    cout << 6+17%3-2<<endl;      //6
    cout << 2+22*(9-1)<<endl;    //46
    cout << (8+7)*2 <<endl;      //30
    cout << (16+7)%2-1<< endl;   //0
    cout << 12/(16-6)<< endl;    //3
    cout << (19-3)*(2+2)/4<<endl; //16

    return 0;
}
```

output:

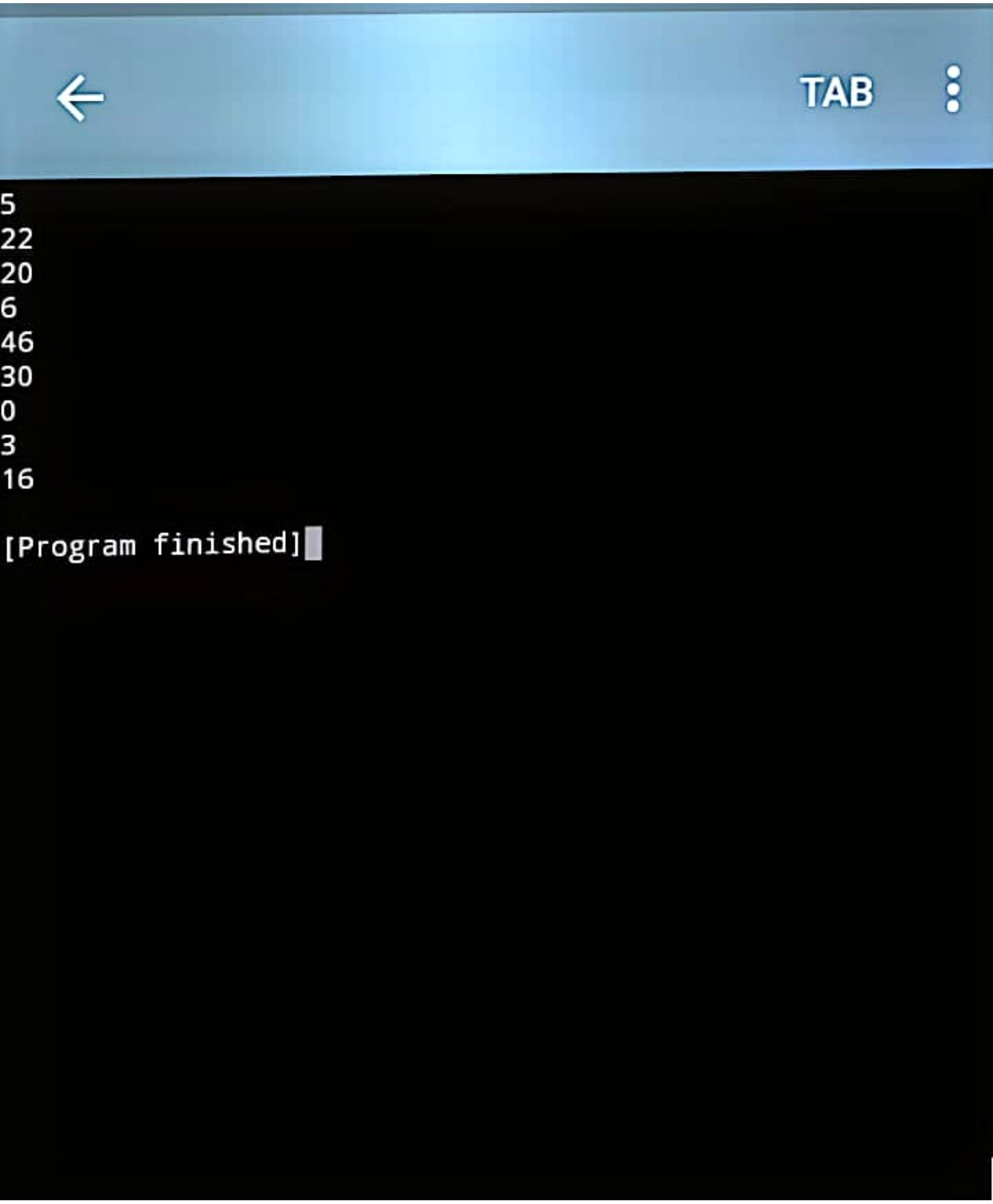
```
5
22
20
6
46
30
0
3
16
```

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new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      cout << 28 / 4 - 2 << endl;           // 5
5      cout << 6 + 12 * 2 - 8 << endl;       // 22
6      cout << 4 + 8 * 2 << endl;            // 20
7      cout << 6 + 17 % 3 - 2 << endl;       // 6
8      cout << 2 + 22 * (9 - 7) << endl;     // 46
9      cout << (8 + 7) * 2 << endl;          // 30
10     cout << (16 + 7) % 2 - 1 << endl;     // 0
11     cout << 12 / (10 - 6) << endl;        // 3
12     cout << (19 - 3) * (2 + 2) / 4 << endl;  //
   16
13     return 0;
14 }
```

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Task: 3

```
#include <iostream>
using namespace std;
int main () {
    int units = 10;
    float mass = 2.5f;
    double weight;
    weight = mass * units;
    cout << "units (int) = " << units << endl;
    cout << "mass (float) = " << mass << endl;
    cout << "weight = mass * units = " << weight << endl << endl;
    cout << "what happens automatically:" << endl;
    cout << "1. unit is promoted from int to float" << endl;
    cout << "2. mass remain float:" << endl;
    cout << "3. mass * units result is float" (2.5 * 10 = 25.0)" << endl;
    cout << "4. that float result is then promoted to double
        for storing in weight" << endl << endl;
    cout << "so correct Answer is: B" << endl;
    return 0;
}
```

output:

```
units (int) = 10
mass (float) = 2.5
weight = mass * units = 25
what happens automatically:
1. unit is promoted from int
   to flot
2. mass remains float
3. mass * units result is float
   (2.5 * 10 = 25.0)
4. That float result is
   then promoted to double
   for storing in weight
so correct answer is
   B.
```

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```cpp
#include <iostream>
using namespace std;

int main() {
    int units = 10;
    float mass = 2.5f;
    double weight;

    // This is the expression from Task 3
    weight = mass * units;

    cout << "units (int) = " << units << endl;
    cout << "mass (float) = " << mass << endl;
    cout << "weight = mass * units = " << weight << endl << endl;

    cout << "What happens automatically:" << endl;
    cout << "1. units is PROMOTED from int to float" << endl;
    cout << "2. mass remains float" << endl;
    cout << "3. mass * units result is float (2.5 * 10 = 25.0)" << endl;
    cout << "4. That float result is then PROMOTED to double for storing in weight" << endl << endl;

    cout << "So Correct Answer is: B" << endl;

    return 0;
}
```

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```
units (int) = 10
mass (float) = 2.5
weight = mass * units = 25

What happens automatically:
1. units is PROMOTED from int to float
2. mass remains float
3. mass * units result is float (2.5 * 10 = 25.0)
4. That float result is then PROMOTED to double for storing
in weight

So Correct Answer is: B

[Program finished]
```

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Task: 4

```
#include <iostream>
using namespace std;
 int main () {
    int a, b = 2;
    float c = 4.2f;
    a = b * c;
        cout << "b = " << b << endl;
        cout << "c = " << c << endl;
        cout << "b * c = " << b * c << endl;
        cout << "value stored in a(int) = " << a << endl;
        return 0;
            }
```

output:

b = 2

c = 4.2

b * c = 8.4

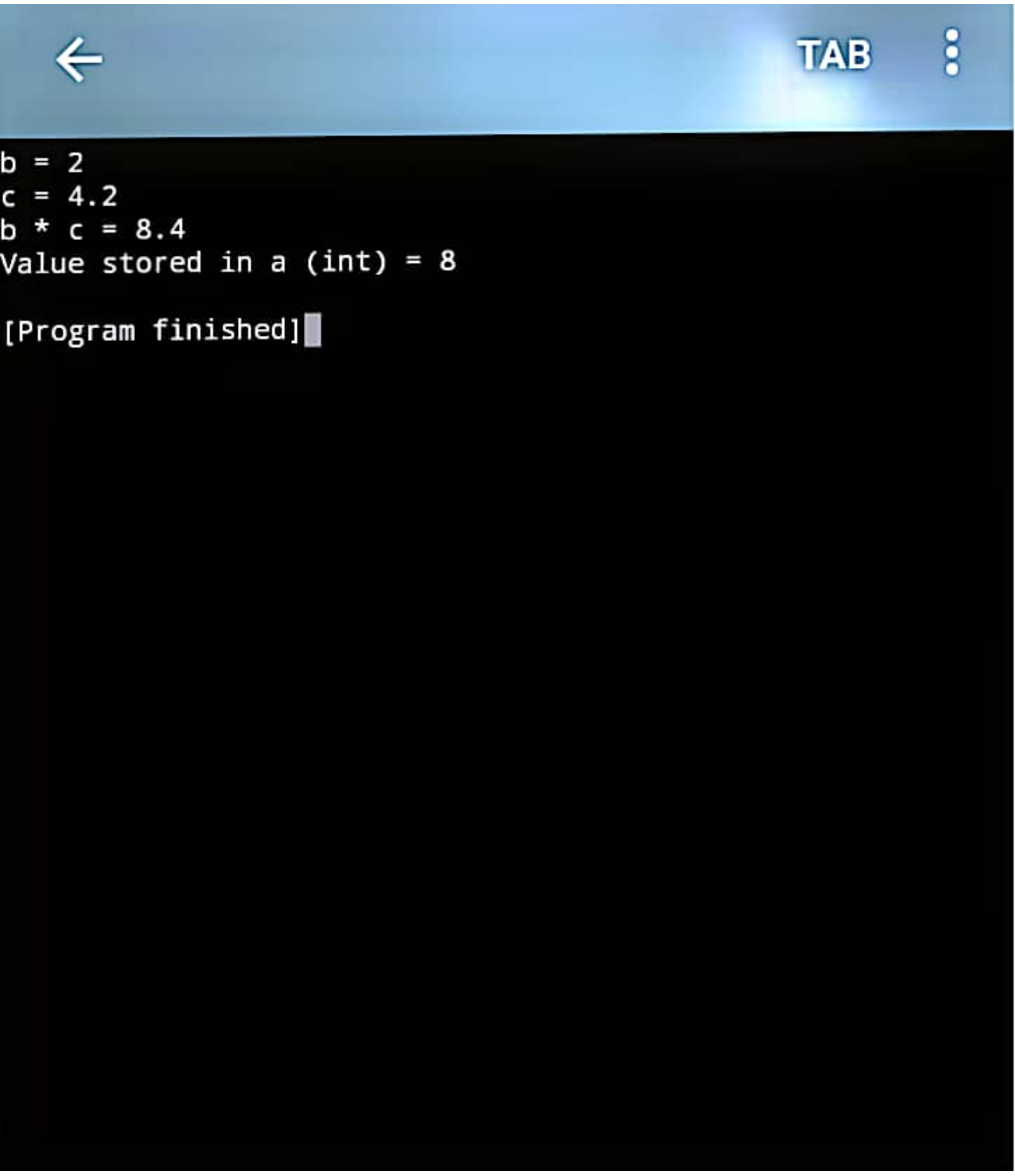
value stored in a (int) = 8

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new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int a, b = 2;
6      float c = 4.2f;
7
8      a = b * c;
9
10     cout << "b = " << b << endl;
11     cout << "c = " << c << endl;
12     cout << "b * c = " << b * c << endl;
13     cout << "Value stored in a (int) = " << a
   << endl;
14
15     return 0;
16 }
```

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Task : 5

```
#include <iostream>
using namespace std;
int main ()
    {
    int qty = 10;
    in salesReps = 3
    double unitsEach;

      // Problem : integer Division
      unitsEach = qty / salesReps;
       cout << "without cast:" << unitsEach<< (wrong,decimal lost)
      " << endl;
      // solution : Type cast
      unitsEach = static_cast<double>(qty) / salesReps;
       cout<< "with cast:" << unitsEach<<"(corred)"<< endl;
      return 0;
        }
```

output :

without cast: 3 (wrong, decimal lost)

with cast : 3.33333 (correct)

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new*

```cpp
#include <iostream>
using namespace std;

int main() {
    int qty = 10;
    int salesReps = 3;
    double unitsEach;

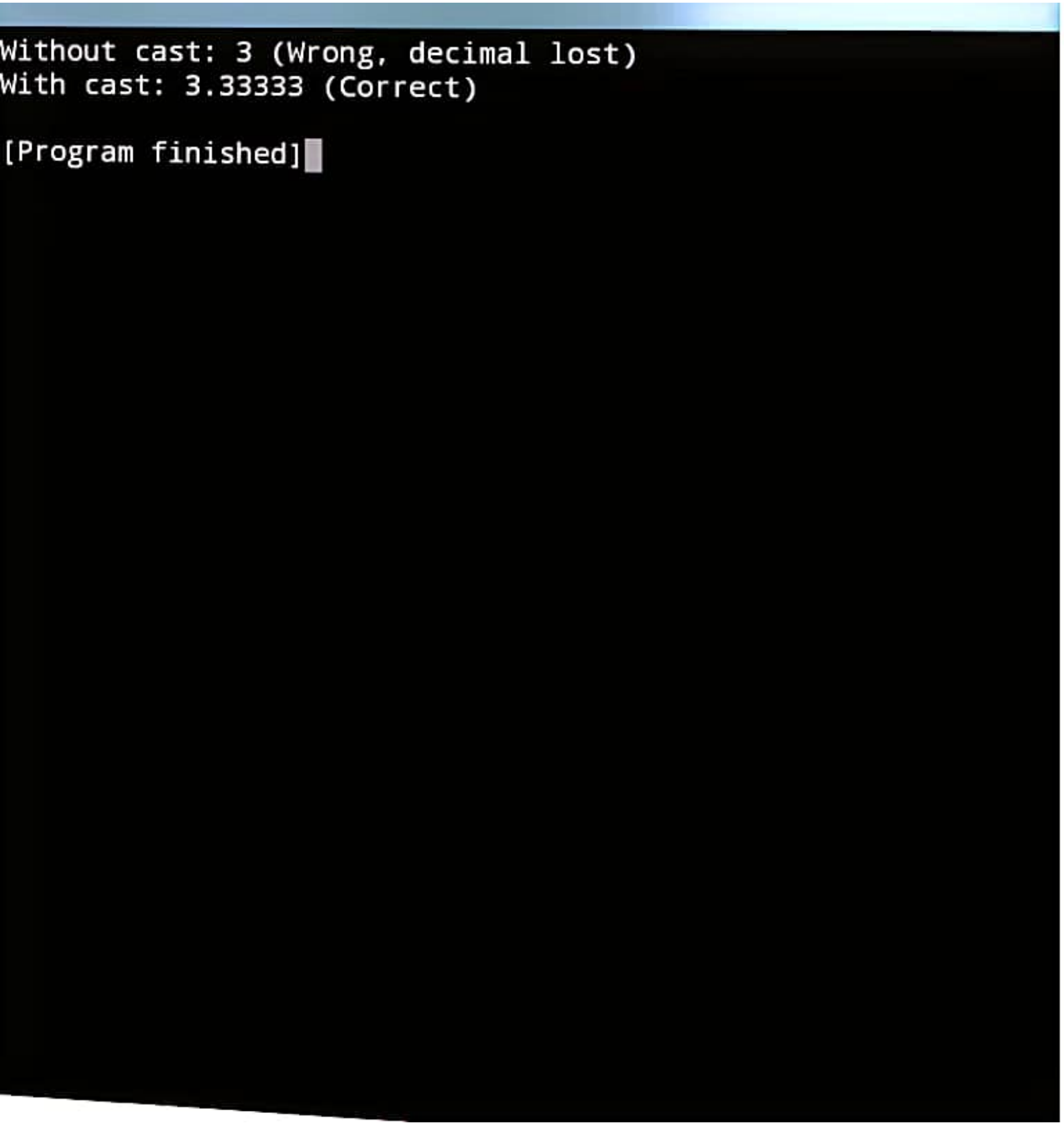
    // Problem: Integer Division
    unitsEach = qty / salesReps;
    cout << "Without cast: " << unitsEach << " (Wrong, decimal lost)" << endl;

    // Solution: Type Cast
    unitsEach = static_cast<double>(qty) / salesReps;
    cout << "With cast: " << unitsEach << " (Correct)" << endl;

    return 0;
}
```

Tab | { | } | : | ; | "

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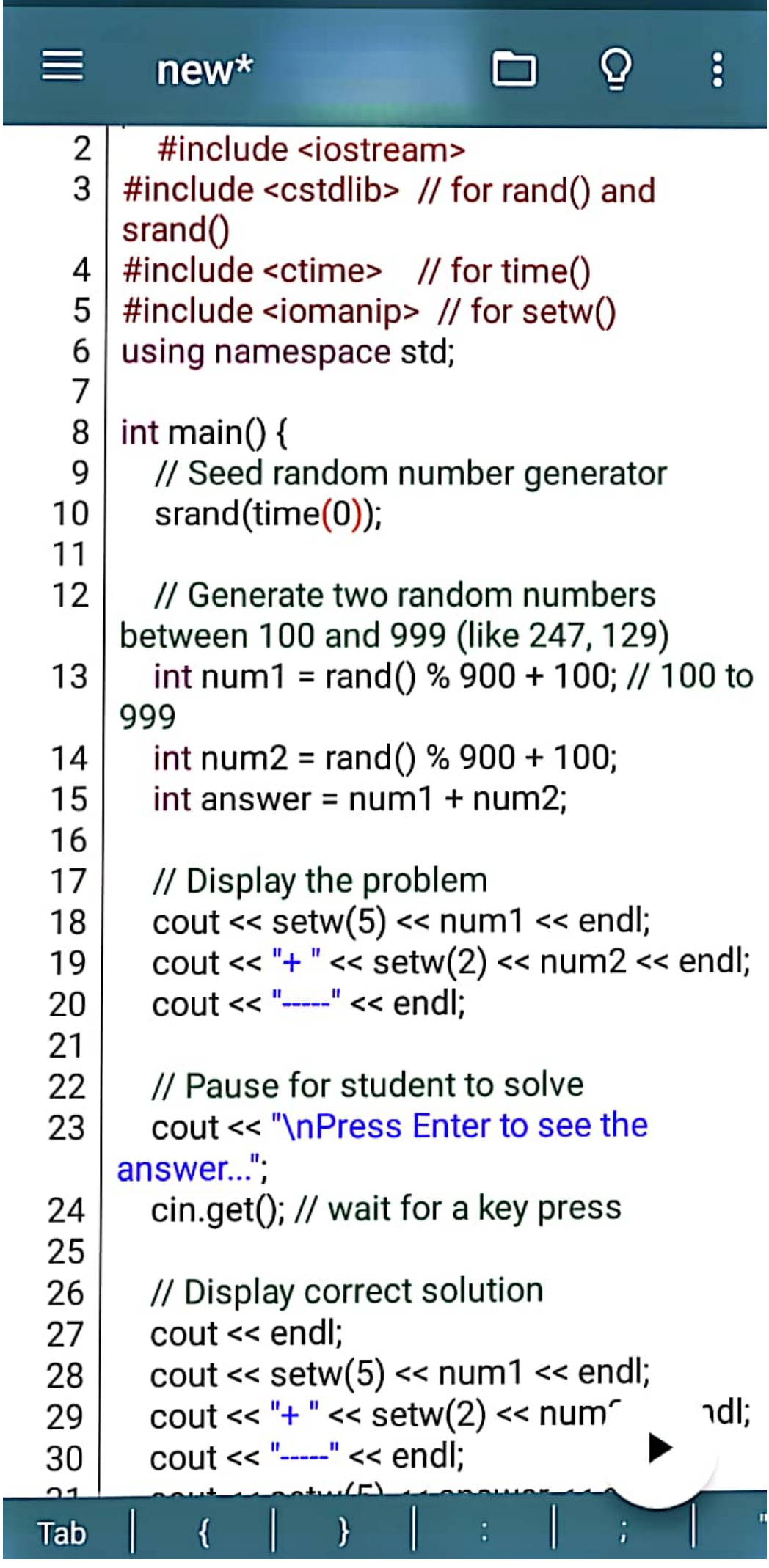
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Task: 6

```
# include< iostream >
# include< ctime >
  using namespace std;

int main()
    {
Srand(time(0));
int num1 = rand()% 900+100;
int num2 = rand()% 900 +100;
Cout << "  "<< num1<<endl;
cout<<"  "<< num2<<endl;
Cout<<"Press Enter to see answer...";
  Cin.get();
Cout<<"  "<<num1<< endl;
Cout<<"  "<<num2<< endl;
Cout<<"  "<<num1 + num2<<endl;
 return 0;
    }
```

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```
  611
+ 885
-----

Press Enter to see the answer...234

  611
+ 885
-----
 1496

[Program finished]
```

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